

test the following ICs successfully: 4000, 4001, 4002, 4011, 4012, 4023, 4025, 4029, 4030, 4049, 4050, 4068, 4069, 4070, 4071, 4072, 4073, 4075, 4077, 4081, 4082, 4093, 5408, 5409, 5411, 5421, 5479, 7266, 7400, 7401, 7402, 7403, 7404, 7405, 7408, 7409, 7410, 7411, 7412, 7414, 7420, 7421, 7427, 7430, 7432, 7473, 7474, 7476, 7478, 7479, 7486, 74132 and 74393.

ICs 4011, 4023, 4029, 4030, 4069, 4093, 7402, 7404, 7414, 7476 and 74393 were tested in EFY lab.

The number of supported ICs can be enhanced with the incorporation of new functions and libraries to the program. As ATmega2560 MCU has 256kB of flash memory, a program for very large number of ICs can be uploaded.

Unlike a typical IC tester, this device provides many useful features to its user. Nokia 5110 display panel and a 15-key keypad were used in this device for designing a suitable user interface, which enables the truth table exhibition possible.

Users can find help on the testing procedure. While entering the inputs (for example, IC number and time interval), they can clear the digits if entered incorrectly and re-enter the correct ones. Truth tables for each individual gate can be paused for better observation and can be skipped for saving time. There is feasibility of reproducing the previous stage (for re-entering the data) without resetting the device.

16MHz of processing speed makes the time response of this IC tester pretty good. No time-lag is observed while accepting data from the keypad and displaying information over the LCD panel. In case of auto-search, this prototype takes about 0.5 seconds on an average to detect the IC. All these features make this device powerful and user-friendly. A comparison of its features and specifications with two other branded IC testers (DICT-02 (Brand: Kitek) and Leaper-1A

TABLE I
COMPARISON WITH TWO OTHER IC TESTERS

Features and specifications	DICT-02 (Brand: Kitek)	Leaper-1A (Brand: Leap)	Author's prototype
Operating voltage	230V AC, 50Hz	9V	7V-12V
Dimensions	26.5cm × 18cm × 6cm	16cm × 11cm × 4.5cm	16cm × 10cm × 2.5cm
USB port	No	No	Yes
Provision for re-programming	No	No	Yes
Maximum pin number	40	24	20 (can be upgraded to 40)
Display unit	9-digit 7-segment display	16×1 LCD	Nokia 5110 LCD (48×84 pixels)
Input method	Alphanumeric keypad	No numeric keypad	5×3 matrix keypad (15 keys) Keys: 0-9, up, down, left, right, ok
Auto-search	Yes	Yes	Yes
Truth table observation	No	No	Shows for all gates
Counting process observation	No	No	Available for counter ICs
Analogue IC checking	Yes	No	Can be upgraded for analogue ICs up to 14 pins

TABLE II
GLOBAL VARIABLES

Variables	Basic operations
int good_gate=0, bad_gate=0;	For holding the information of the working condition of each gate
int i=0, x=1, ret=1;	Used in some loop conditions
char key;	Holds the character entered through the keypad
char str[5]; char str1[5]="---"; char str2[5]="---";	Holds some strings for display purposes
nokia_lcd my(9,10,11,12,13);	Specifies the pins connecting the LCD panel
matrix_keypad inp(28,26,24,22,23,29,27,25);	Specifies the rows and columns connecting the keypad

(Brand: Leap)) is shown in Table I.

Software. A brief note on the coding is shown in Table II with different uses of variables and in Table III with uses of important functions.

Construction and testing

An actual-size, single-side PCB of the transmitter unit is shown in Fig. 4 and its component layout in Fig. 5. One can use this PCB as Arduino shield with Arduino Mega ADK board using 12V power supply at CON2. Otherwise, Arduino Mega ADK board can be interfaced with the PCB using cable con-

nectors and power supply from 12V/1-amp adaptor.

As shown in block diagram (Fig. 2), the MCU is interfaced through an LCD, a keypad and an IC ZIF socket. The flow chart corresponding to the basic working process is shown in Fig. 6. As shown in the flow chart, this prototype is equipped with two methods for checking a particular IC. Both of these are elaborated below.

As different ICs come with their own specifications, checking process for each may vary. Here, we take an example of a common NAND gate